

Turn on, Tune in, Observe with Karl Friston and Markus Müller

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method: auto_caption

hello and welcome to the Greek Minds series I'm super excited in this series we have not one but two exceptional guests I'm delighted to welcome our first guest Professor Carl Friston a leading neuroscientist known for his groundbreaking work in brain Imaging and his development of the free energy principle on the other hand Dr Marcus Mueller a physicist whose Academy Journey step in theoretical physics and Mathematics he's worked delve into the Intriguing Crossroads of information Theory and quantum mechanics today we are stepping into the realm of theoretical neuroscience and quantum mechanics as we dive into the physics of the first person perspective okay thanks very much so let me say in advance thanks so much for your interest and and your patience your interest in all of this um let me begin with it with a disclaimer so what I talk about today is very speculative um and probably to a large extent wrong yeah it's also very abstract and counter-intuitive so that's makes it a bit challenging for me to to explain it and to like depict what it has to say what I can promise though is um that it is a consistent approach it's I think well motivated um if it's true then it solves a couple of interesting questions but if you believe well I don't possibly think that the world could be that crazy then that's completely fine then you can just see this like one hour of like fun science fiction perhaps yeah so indeed um I'm usually working on other things also some interesting overlap apps with Professor for instance work on free energy so in a very different context um now what I'm talking about today is in some sense a very concrete a concrete proposal for a very different or unusual picture of the world in which the first person comes first in some sense um but I should also say that this has almost nothing to say about concrete everyday questions like such as like how concretely does the brain work right so it's also not a you know Theory of Everything or something like that because it has nothing to say about most physics questions that we're interested in um just to make this clear um here's a comparison so suppose you're a proponent of general relativity you know and then you bring

forward this very very popular thought experiment of an observer in an elevator so this is for experimented if you're in an elevator and you don't have Windows to watch what's going on outside you will never know whether you're floating in outer space or whether you're free falling on earth right um now suppose that they bring you now together in a podcast with a very renowned biologist who's really an expert on the biological details of how the sense of balance works then it may be quite hard for you and the biologist to have a lot to say to it say to say something to each other that would benefit the specific questions that that you're asking um I I feel a bit similar so I feel like I don't really have anything concrete to say or directly to say about how the brain works um or what the neural correlates for example of Consciousness are or something like that I also suspect that um what I'm proposing the abstract core of it is Manifest the indifference to many of the specifics of how humans or other instances of observers um just happen to be composed and and what they look like right like the specifics of the world that we happen to observe nonetheless I I'm looking forward to this discussion so perhaps there is some common ground in the end I definitely have to say that I found some of Carl for instance ideas were interesting for more unpopular scientific levels so sometimes you know when I read on blogs about trapped priors is explanations of some psychiatric conditions also explained with free energy principles and I find this very interesting but it doesn't directly touch my work now I have three things to say basically first is why should we have a thing like how is it done in general and what does it predict but before going there let me ask back like how much time I should take or whether there are any any questions or comments so far before I go to these questions you should take as long as it as it uh you you feel you need I for me I'm looking forward to this this is going to be a learning experience uh for me um just a little just just to relax you a little bit I I spent the weekend reading about you I had to read the proof of George's new book and he clearly spent some time in conversation with you I'm looking forward to hearing the um a summary of the ideas that you were discussing with Georgia as I'm sure the audience are so you take as long as you want okay that's great thank you so much um good so as I've had three questions why how and what does it tell you so so first why so what are the kinds of problems that this approach tries to solve so um consider the following question the question is what happens to me next yeah or what should I expect to see next um now the goal of my Approach is to try and give and in principle answer to this questions in all situations including very exotic ones so let me begin with a simple standard situation like when I will ask what what will I see next so suppose I go to the laboratory here at at The Institute I see a particular arrangement of mirrors and of lasers and other equipment

on on the table um and then what I can ask is well for example should I expect to hear the detector click now certainly I can use physics in particular quantum theory to compute a prediction for example a probability of hearing a detector click so this is what we usually do in physics this is like the standard methodology we say we remodel the world or the relevant part of it and it's Evolution then we locate ourselves inside of it and then we calculate what that kind of piece of matter is what's happening to that right that's essentially what we do even if we don't spell it out always um but then there are these exotic situations where this prescription doesn't quite work um for example the computer simulation of Agents so somebody promises to scan you um to whatever detail um and simulate a copy on a computer for example um should you well what should you expect should you expect to to make observations that correspond to the simulated agent or not and perhaps you want to put a probability on that I claim that physics is in principle silent about this question and then there are these duplication scenarios put forward by philosophers like parthit which are something like these you know Star Trek science fiction scenarios you enter this teleporter you know you're you're dissolve and then a copy gets built up on the planet where you want to travel but then you may have a malfunction and suddenly if two copies appear or more than two copies with slightly different properties how should you think of that say before it happens you know it will happen What should you expect to happen to you like you know um in physics in quantum theory in particular there's a puzzle that appeared that's called weakness friend the weakness friend Paradox which somewhat essentially asks how you should think of being put in superposition and whether what you predict to observe is somewhat consistent with what other agents might predict to observe something final example would be what's also appeared in physics the the boltzmann brain problem I will explain this a bit more broadly because I will pack it back to it so this is in a nutshell so I apologize to all the fellow cosmologists to know this much better than me but the main idea is very simple suppose the universe is very large perhaps your model predicts that Eternal inflation or some other model predicts that it's perhaps combinatorially large and then just due to thermodynamic augmented fluctuations you know that a lot of stuff must just go on randomly out there in particular in some of these models you expect that there are many copies of you just appear due to random chance statistically they would be perfect copies of you and your memories but in the next moments you know these copies would somehow make a very strange observations like what the hell is going on and disappear yeah it's a very colorful depiction the cosmologists really want to think about this because they might say well if your model predicts that then maybe your model is

sick and you want to rule it out but again the question appears how should you think of that in particular if an observer in such a universe as we think perhaps what should we expect in the next moment to happen to you usual Planet business or strange fluctuation experience right it seems like in all these cases the standard methodology does not directly apply it's at least unclear how to use it so you need something else so so that's the why now um I think it's just a worthwhile goal to try and have an approach that attacks these questions uniformly and says here's a method how you can in principle work it out good now how could you do it perhaps um um well how can you can you build now a theory that would allow you to do this there are several options yeah others you know I'm not saying I found the only one it's not even a full Theory but the the main idea is when you say like if I cannot apply this methodology like here's the world it evolves here I am then gets one step back let's get one step back um and not talk about the world let's let's first talk about you yeah whatever makes you you so mathematically you would want to say that something like the Observer state so that's whatever it is that is you right now basically a snapshot perhaps of will not easily think of all the the relevant content of your brain including memories observations whatever will in the end Define you um we would think that perhaps experience and Consciousness and all that super beans on that somehow but that's not necessary that's not important for what I'm going to say right so your state doesn't have to be conscious or anything so think of you during a surgery it doesn't need to be human or biological think of just some pattern somewhere that causes what happens to that thing let's talk about the Observer state in itself and let's postulate a law that says if you are in that particular Observer State now you will be subjectively in another state next with some probability yeah um ideally that's it you just do that you try to guess a well-motivated probability law and then you work out the consequences and compare it to what what we observe yeah now certainly what you want for that to be successful with your demand from this is that it should agree with the physical predictions in the standard situations at least in principle even though it's maybe hard to apply in practice so you know you want to see the situation where you go again to the laboratory at The Institute with a description of lasers and mirrors and your state which includes all the memory and whatnot things you're not aware of that make up you you want to be able to prove that this approach in principle gives probabilistic predictions that are consistent with those that physics would give you um if that's the case then you might trust the approach enough to apply it also in these exotic situations that's that's the hope and that's the that's the goal now you can ask well but what could this probability law be what would be a good choice and then I think a good like

well-motivated choice would be to use something like a reverse principle principle so it's like the inverse of David Lewis principle principle and what it would say is that the the actual chances that you postulate what happens to you what are the probabilities um they they should coincide with the epistemic predictions of maximally rational predictors so what does it mean like more concretely well I suppose I'm given you your state right now yeah whatever it is um I put it on the table I cannot really do that but suppose I give it to a team of scientists they are very well funded they have all the time in the world and they look at this string of data right now you can ask what what they unlike what they could probably do is they could work out that oh that looks a bit like here looks like a recording of some Digital Image maybe they there seems to be a dump of the brain of some being in a three-dimensional world and here it can infer that stuff was moving in some ways just from the recorded data you can puzzle together a lot of facts about the world where this came from and then you can make a prediction and say well if that's the case then this is what this guy should see next all right so this is um a version of universal induction and to formalize that you can use algorithmic information Theory so algorithmic information theory is a part of information theory that gives you different options to formalize this in in detail what you end up with essentially say my next state is more probable if it's more compressible given the current one so in some sense if the next statement looks more business as usual or more like a natural continuation of the current one in a precise mathematical sense okay so that's that's the main idea what you would want to to formalize yeah um let me point out there are many options how you could do it in detail for example also how you choose how to model Observer States mathematically and so on right but it's also very simple and you want to stick to it and not postulate other things and see what things it can reproduce that we know about our world so then if you do that then what so there are questions what what does it actually predict or explain perhaps um one thing you get is automatically essentially is you get consistency with physics in principle assuming that some version of the church during thesis holds for our physical world the church during thesis will tell you for example that if you build an experiment and you feed an exhaustive description of the experiment into a computer the computer can calculate probabilities of future outcomes for you so there's an algorithm in principle that can that you can use to predict if this is the case then you can prove that in the long run the probability law that I've told you will coincide with the actual physical predictions so that's consistency with with physics um in this view however you are not fundamentally actually embedded into some physical world instead the approach predicts that in the long run U.S states will look as if you were embedded into some larger

system yeah that system is computable by a short program on a probabilistic turing machine [Music] and you can now call this your external world if you want so tells you that there exists this this model where you can imagine that you that you're part of that thing so what the probabilities that you see are marginal probabilities of this larger process and that would mechanistically explain what you see so this is a sort of emergent notion of an external world importantly this is not a theory of everything so a priori this approach predicts almost no properties of that except that it's computable that it's probabilistic and that has probably simple laws even though the actual realizations of the data you see are probably complex now this seems to be an agreement with what we see but there's no claim that this could actually give you the laws of physics or you know it would be a theory of everything of that sort now at this point you might also understand already how it how it solves the boltzmann brain problem what what it has to say about that so again let me rephrase it quickly so suppose this world around you that you infer that you think sits there and you know is consistent with the observations suppose that has that property that it's very large and that you should expect that there are many copies of you sitting around now the first intuitive question is a human that you want to ask is now given all these copies am I the real guy on this planet or am I one of the copies yeah that's what he would want to ask intuitively now now my Approach will tell you this is not even a meaningful question so there's no property of you that will tell you one you're one and not the others right you adjust your Observer State your pattern right and that's the same pattern here and in the in the fluctuation out there it's like the number six is just the number six whether it's these six eggs in the basket on my desk or the six soccer balls in the room of my little son um however here's a second question that you can ask and you can ask well good maybe that doesn't make sense but what should I expect in the next moment to see should I expect to see business as usual on Earth what I should expect to see something strange that would happen to these fluctuations and then the answer would be this is yes this is a meaningful question and you should expect to see business as usual continue why is that well that would give you observations that are more compressible given your previous state algorithmically okay so this would also be bad news for the cosmologists so in some sense huh some cosmologists want to say hey fellow cosmologists your model predicts so many botsman brains then it must be wrong because it's not consistent with our observations but in in my Approach um you cannot do that right um even if there are many copies the sheer number of the copies would not have any bearing on what you should expect to happen to you and on the chances of what will happen to you finally this is like the most counter-intuitive aspect of it

you do get an emergent notion of objectivity um and this is like the last thing before I finish my long monologue and it's perhaps the the most science fiction and fun part here's what happens to me right so I I have these observations I'm in the state and to me it looks very much if there was this world around me and I can point at something over there um that thing or that that that seems to to me like another Observer has yeah carries a lot of data I would call this professor Carl friston right and so I can predict that Tomorrow there's an almost hundred percent chance that I will see Professor friston see the sunrise right so I can work out the probability that I will see this other person see something all I need for that are the laws of physics I was like piece of matter that records data does it like to the camera eyes look at the sun yes or no good now you can ask but what if you are actually Professor friston right so if you are that Observer State that's represented in this data over there um you can also like my Approach claims that then you should believe your chance of seeing the sun rise is just given by this Universal induction probability by algorithmic probability and then you can prove a theorem that these two probabilities are very close to each other as long as what I'm pointing to carries enough information and is complex enough so this will tell me that my world is also Professor for instance world and that appears also to other things that are point to in my world even if they are not biological like the data on my hard drive for example or other things that carry a lot of data about the external world it would not however apply to simpler things and then we would have the notion where you know the the notion of objectivity breaks down in some sense and you get some something I call the probabilistic zombie so this would be situations where me seeing something see something has a different probability than that other thing actually sees something and that's totally counter-intuitive and that's that's a noble like phenomenon that's that's predicted by approaches like mine good so so that was kind of an uh a summary of my proposal um as I said it's it's not a full Theory yet so I have the general idea and perhaps you can see some of the ideas worked out and explain how they should work I have a concrete model worked out something like a bit model where you just model The Observer State as a string of bits that grows one bit every time that make a new observation and there I can prove rigorously some of the things I claimed but certainly actual observers forget something sometimes fundamentally so that's why the rigorous version that I put forward in this paper here certainly does not apply yet um and there's a lot to be tried and improved and you may try out different versions you might also try to falsify it when you say here's statistical predictions perhaps too strong fluctuations expected on the laws of physics we don't see them then we can put the theory in the garbage can also right so so that's also

something to be worked out but yeah I'll stop here and thanks thanks for for listening for such a long time for your attention love it doctor thank you very much that was intriguing there's so much um that one could pursue there um I was intrigued by this notion of probabilistic zombies so I I'm probably gonna ask you to unpack that a little bit more but um just to celebrate points of contact um you know I'm now pretending to be the biologist who knows how the sense of balance works and trying to try to make sense of uh your more um fundamental and foundational physics perspective but having said that a lot of what you said um makes entire sense um when viewed through the lens of at least a bio mathematician um you know um so I just wanted to sort of highlight the points of contact which I understood um as a Prelude to then going back and looking at some of the more outrageous um or counter-intuitive implications of the way that you're actually deploying this particular Theory um so a whole bunch of um consilient observations um that come from your formulation um one way of reading your notion of um the natural way to predict what will happen next in terms of the um the compressibility and the um the minimization of algorithmic complex complexity that that that is something which um I you know I can completely subscribe to or when I say I uh somebody committed to understanding things in terms of the the free energy principle would absolutely be there uh completely and indeed there are I think two routes to um motivating our understanding the foundational role of or um two roots to understanding processes that look as if they're optimizing something and the thing that they are optimizing is some variational bound on marginal likelihood or um model evidence uh for me that would be the variation of free energy and one route would be some of the Richard Feynman route uh from Quantum electrodynamics and sort of thinking about um how you how material things things like you and me um from a biologist's point of view um would actually make sense of their sensory data in exactly the same way that Richard Feynman was trying to work out um you know how he could make sense of distributions of different small particles pursuing different paths and trying to you know derive the probability distributions over those paths which is an impossible marginalization problem or integration problem but can be solved by converting it into an optimization problem so now if I'm solving the same kind of inference problems or perceptual measurement problems or Observer problems then it you know if I use that kind of trick I will certainly look as though I'm optimizing something which is bound on the free energy and uh and that would be I think the sort of um the the story or the narrative um that people like David McKay and Jeffrey Hinton would use in terms of appealing to uh um that that pathic reform formulation um as the Genesis of free energy bounds you know evidence lower

backgrounds in machine learning but there's another route um and I think it's probably one much closer to your formulation which uh I fondly think of as a Russian route so the final would be the American route but there's another that starts with your algorithmic complexity and formalograph and you know and provide that right through so often induction and Universal computation um actually interestingly picked up by people like David McKay um in crypto analysis that that that they're exactly the same um maximization of marginal likelihood and the minimization of free energy um sorry these things are exactly the same as basically representing things in a in a compressed fashion that minimize the algorithmic complexity um and interestingly um if you look at uh what current biologists are certain neurobiologists like to think of as a way of implementing this compression um what it would look like in the brain you'd end up with something called predicted coding which you you know I'm sure you're familiar with and of course predictive coding was certainly in its linear form was something that engineers invented in the 1950s do compress sound files to minimize algorithmic complexity of the of the encoding of large sound files so to my mind there's a wonderful sort of if you like convergence on exactly the same quantity and indeed people like Jurgen Smith Hoover have been sort of you know shouting about um all you need is complexity minimization to explain everything but for for many many years now so I just wanted to um emphasize this notion that there is I think a natural law out there that would be under it by um a compression uh a a drive to efficiency a probabilistic um maximizing the efficiency of something that you had observed if if you that thing you observed left its impression upon you and your state you know that you were describing all the states so constitute me and my brain and all my internal Dynamics um then I think that you know the argument that that natural law would rest upon this you call it Universal induction is that what you is that the phrase that you used yes I used this phrase yes yeah is that related to Universal computation then is um as read by your by the by the sort of people who inherit from either solemn off for indeed have you come across the work of uh Wallace from from Melbourne University um in Australia uh so it is solomonov's prior by the way I haven't come across Wallace's work which which what's the first name I'm afraid I can't remember because he was always referred to as Professor Wallace but I do I remember visiting Melbourne once um and they had a philosophy slash maths group um that that um and then so what Professor Wallace was was basically the the head of department and had died um a few years prior to me visiting but he was certainly outside Russia the real champion of this sort of uh algorithmic complexity view of um of stuff self-organization through to you know efficient coding through to Universal computation and

wrote some seminal papers that they were very proud of so so I I always try and reference this gentleman professor professor I hope I've got the name right I'll email you afterwards um in relation to that because you know I think that the Australians felt they've been left out of the the big arguments with the you know the Canadians with with Jeffrey Hinton and the like and the you know some of the Americans and but I just use him as another point of reference that somebody you know outside Russia saw this um as a your fundamental way a universal way um of understanding computational or physics as computation or information processing um you know from the point of view of com of complexity uh and just to bring this point of contact to closure uh you know that um literally underwrites uh from the point of view of crypto analysis uh the kind of variational free energy that I you know mean when I talk about minimizing free energy um when I when I think about this and certainly an explanation to students when taking a perspective on variational free energy as an Evidence Bound in statistics what I generally say is is that you can just rewrite that marginal likelihood or um the variation of the energy in physics would be the negative uh log of that marginal likelihood as a linear mixture of the accuracy and the complexity where the complexity is a callback KL Divergence between your prior and posterior so before and after getting the observation um the impression of that observation on you your Observer State um can be scored in continuous formulations so this would be your appropriate for sort of continuous State space models as the KL Divergence between um the the prior and the posters are literally scoring probabilistically the degree to which that observation changed your mind uh yeah you know that fluctuation what you know what how is how has it moved you from T to t plus one and um that's the complexity um and of course because the log evidence or the log marginal likelihood is equal to accuracy minus complexity if you're trying to maximize as an every good statistician should the evidence for her hypotheses um then you are going to try to maximize the accuracy of your account of this observation but under the conclusional constraint you're trying to minimize the complexity uh and it is that link I think which is fundamental and sort of ties you to the universal induction and Universal computation that's the compression so you don't want to change your mind too much because every you know if you change your mind and use up degrees of freedom you're now incurring a complexity cost which is exactly uh what you don't want to do so I normally read that sort of Occam's razor aspect um which I think you could also read in terms of James's maximum entry principle um or even Jeffrey's prize and statistics I I read that as basically um get it getting the right priors to enabling you to compress everything so you don't need to move away from your prize if

you can compress this information and understand it and rewrite it or explain it in terms of good prize you've already got then your posterior doesn't move very much and you've now got the most efficient compressed representation I I'm not not sure about the maths and I'm sure you'll be much more fluent in this than I am I remember looking at into it once and trying to use the law of large numbers to connect this um statistical measure complexity to the algorithmic complexity but at least heuristically I I just look at these things as just two ways of looking at the same phenomena one from the point of view of continuous State spaces and one from the point of view of sort of you know discrete bits or you know of a kind that you could apply algorithmic complexity to have you come across that sort of statistical complexity thing before the uh yes yes indeed so I'm glad you you say this um so certainly a lot of what you say is corroborated by a couple of theorems that relate for example chronograph complexity and entropy or you know um conditional comographer of complexity and coolback Library Divergence so essentially the expected value of chronograph complexity is entropy um so this is why you intuition that you mentioned like you know how surprised Am I by the new observation giving the previous one is mentioned by the Callback Library Divergence can be interpreted as some kind of expectation of a conditional algorithmic um complexity except certainly that you're in the continuous regime no so um what I said is true for the for the discrete one but but surely this connection can be made and this is something I I definitely find interesting because I'm indeed looking at alternative ways to model Observer States right to model probability distribution um in particular ones where I don't have to add a bit at a time I know I wonder whether I should you know study in more detail how you actually formalized it um I haven't enough right but yeah go ahead well yeah I would certainly encourage you to do that because um it is seldom the case that I meet somebody who is uh fluent in both the continuous sort of um uh um data analysis kind of computational complexity and the you know the inherent information geometries and continuous State spaces I think you know things like um vicious information uh metrics and information geometry and you know how that relates and how thinking about this um complexity measure as a KL Divergence moving from your post device to from your prior to your poster is literally a movement on a statistical manifold that has an informational lens that whose metric is efficient information so the whole bunch of beautiful maths but it's all in continuous State spaces I if if you could understand and become fluent in that to the extent you could join the dots with the with the pure algorithmic stuff I think that would be a great service I'm sure these are exactly the same stories written in a slightly different rhetoric from the point of view of different kind of mechanics um uh you

know and I'm not sufficiently well versed in the algorithmic uh complexity and Universal computation to really sort of say these are basically the same kind of story if you could do that I think that would be uh would be you know you're very very useful and you you'd make a lot of threads or at least allow people to to see each other's perspective so I just I I just really encourage that if you know if it can be done and don't forget to cite this this Australian gentleman Wallace is it Professor is it Chris or Mark that is two one of it is called Chris uh Chris Wallace is into information science is that could be him uh that doesn't mean he used to be it um I'm pretty sure it's Monash University in uh Melbourne in uh it's Mark Wallace I think Mark Wallace it's Mark Wallace yeah right yeah and you know he he would have been writing I think in the in the sort of uh 80s and late 90s if I remember correctly so I'll do my research I'll send you my favorite citations that sort of captured the essence of this you know the things that we've been talking but that is really interesting and I thank you so much for pointing me to this I do have a somewhat question for you of technical curiosity so um are there results that say something like if the environment if the world is of that on that kind then I obtain a guarantee um that the free energy minimization will be successful for survival for so I guess so for algorithmic complexity for example it would be a version of church during thesis so you need some in principle computability but probably you would also be able to say something about um for the statistical version like probably the environment shouldn't be too unpredictable in principle or something like this I see um yes I I'm afraid that there isn't you know well um and I'm probably not the best person to ask I mean basically you know because my background is more sort of intuitive physics and and yes mathematics but um there are certainly um a lot of Notions that would speak to that um that basic outcome that if you are um embedded uh in some meaningful way but individuated from um a particular environment external media some some Niche um then there would have to be um a non-equilibrium steady state exchange um between you and the Observer uh and the thing generating observations I know you may not like that because your health thesis isn't there's nothing generating observations in my world but with slightly less uh less philosophical about it you you assume that there is there is some World there and you've got some um Observer uh and they are individuated in some well-defined sense so that um the observations that are doing the updating and the updating is in the direction of um being maximally compressible or minimally complex uh under the constraint of it's still providing a a good representation or accurate explanation for um just Open brackets you know if you could show the equivalence you've also now got a a con a duality with James's maximum entry principle if you read the

maximum entropy about of the probabilistic description of what caused your measurement your observation so that you know the constraint just is um well one way of reading this is that the constraint um is supplied by or at least scored by the accuracy of um you know of of your measurements it's you'd have to move certain terms around um but put simply you know the free energy formalism can also be read allergens in the sentence that you've got an entropy of some belief distribution encoded by your Observer States um that um has to be maximized in the spirit of Occam's razor you don't want to commit to a particular explanation so it has to be simple in the sense it has to be liberal and non-committal but you've got a constraint and that constraint can be written down in terms of lagrangians or LaGrange multiplies or uh or specifically in the for the point of view of the free energy principle um a um a log a negative log probability um of particular observations and indeed the other the the external States causing them um sorry I'm just I'm just uh adding a little bit of momentum to my um and uh encouragement to you to to actually you know join all of these dots because if you can join it to the free energy principle the statistical you know the basis statistician's use of of um uh variational um free energy as a you know as an objective function to get good models of things to make sense in the best kind of way of observations you've also got an easy um point of connection with um with um Max eventually maximum caliper uh um principles from James as well so you could tie it everything together but sorry to come back to your question the so in that respect um then you can um in fact the answer to your question really is the the the the race on Tetra or the free energy principle that if you have um a system that is um in some characteristic States on that system so the Observer can be separated from the thing that's providing the observations then just by the existence either of a non-equilibrium steady state an open exchange so these are all about open systems then um just by either the existence of that non-equilibrium satisfied solution or the existence of uh Dynamics um at any you know for a suitably long period of time then you can always um explain the change in the state of the Observer um in continuous State space formulations as a gradient flow on the the other the the variation for energy that has this complexity uh computational complexity part of it so the answer is yes possibly their way around if you are in non-equilibrium steady state exchange with your environment um another way of reading that would be you know if you um add events some kind of generalized synchronization of chaos if you want to put some random fluctuations in in play then there will always be a description of your next state or your current Dynamics as a gradient flow on the same quantity that hopefully we share a commitment to uh that has this sort of compression uh compressibility aspect to its efficiency aspect uh to it um

so that's certainly true uh early versions of that which you know would be another nice thing to sort of draw into the fold of things that that are all if you like um pointing the same direction inheritance on cybernetic so this is the work of people like Ross Ashby um and Colin um so things like the good regulator theorem and the law of requisite variety could in one sense be read as answers to your question so uh have you come across those those early Notions no I'm not at all familiar with this oh you'd like that um and he also um was able to prove stuff I never really I had I tried to read the papers I never really understood the proofs um but yeah I think you would like this so that this is um this is Circa uh you know probably um 1950s possibly later 70s um but um but certainly the early work of um so the good regulator theorem just says that any Observer that it lives in a controllable world must be a model of that world um so and the law of requisite variety can also be read as the degrees of freedom of the Observer the observer's state must be at least those um or have a cardinality of the environment that is being controlled so again this notion of being and it's a notion I think was implicit in your opening comments about you know where should I look next so it's all about actively controlling not controlling but engaging with a world so you're open to something that you're observing so you can control your observations by where do I look over here over there so from the point of view of the good regulator theorem um is assuming or they were assuming um that there is an observer um there are things out there that could be controlled you know you know the Observer doesn't know you don't necessarily need to know the structure of what those things are there just needs to have enough degrees of freedom on the inside the internal bulk um to match the you know those things that can be controlled out there and then it is provingly the case that if you have the open exchange with the world then the internal um the internal Dynamics The Observer states must be readable as or be a good model of the uh of what's going on out there and it may be that the proofs do actually um sort of your portend the later derivations um that you know eventually several decades later led to the universal computation and as I understand it Universal Universal induction um but you'd have to go back to the yeah I think to that very early literature um so that would be one other way of saying that yo if I exist in a world and I can observe that world under the caveat I have to uh control the um the way that I solicit those observations I have to actually you know have some reciprocal exchange with that world then yes the you know there can be no other solution to that than me being a good model of that world of course what is being a good model mean well it means I've got a compressed representation of the causes of my Sensations and what does that mean if I'm um if you're in Universal time well it just means that my States

will be those that maximize the model evidence in a continuous formulation not some model evidence it's exactly what we're saying before it's a thing that's bounded it is just the mixture of the accuracy and and the complexity so I think it all it all does fit together um I know I repeat it will be interesting if I remember if I remember the proofs of the co-regulator theorem um although they may be a little bit heuristic from for your tastes given you know the formalisms that ensued they may have been articulated in the kind of language which which you you you would recognize I said it's an elemental umesque information theoretic um yes indeed I'm actually intrigued by a description I should say um in many cases what I used are theorems that as theorems can be read either way so that could be applied like thought of being applied in situations where I think of an observer that is actually embedded in a fundamental way or use the theorem and turn it around and use it in my Approach so to get one step back I'm just interested in in this for like what you can prove that in generality and what he said intrigued me because this could regulator theorem when you say well the Observer has to have enough degrees of freedom for the world somehow The Observer states must contain a good model for the world in order to have the ensuing sustainability of that Observer that's mathematically really similar to my condition for not being a probabilistic zombie Ah that's essentially the same condition so the Observer state must contain enough information about the part of the external world that affects affects them and if so then it's not a probabilistic zombie and then the probabilistic predictions that you would put forward if you just describe that world would match the probabilistic predictions that you would get if you applied it directly to your server state so I'm intrigued by these proposals because they seem disappointed some very deep mathematical underpinnings that perhaps can then also be used to make a more conceptual relation to this um more statistical approach that you're working with I definitely want to pursue this so I thank you already for for pointing me to this and we've happily got back to the probabilistic zombies so that's that was a big clear can you just speak more about the problem that was intriguing the probabilistic zombie yes so in my Approach the notion of objective reality is not assumed and built into the formalism as you would typically do but it's it's the content of a theorem that you prove right um and certainly every theorem has its premises and it isn't that premises you get that conclusion and the the premise of of this theorem is has to do with that other thing that you're pointing though that the interpret as another Observer containing enough information about your world um right so you can quantify this in terms of chronograph complexity essentially um and of the world I really mean essentially the laws and not the actual realization um somehow when when you

when I pointed your brain and I I'm I think that we we have objective descriptions or what I think my probabilistic predictions are for what happens to you should agree with yours then this has to do with in my Approach with saying well the comogra of complexity of your state over this is much larger essentially even the comograph complexity of of the laws of the worlds that affect you at the regime where you live just the laws not the actual data of the air molecules or so that would be a high complexity but the laws by which they move um so intuitively I would say that this applies to you know certainly all other things we would intuitively call Observers it doesn't apply to um things that are point to that have very slow small information content and it also applies in situations where so how should I say so like computer simulation of Agents you can simulate a simulated being in a simulated world on a computer um may have large information about its simulated world but not about ours about our world yeah that's why in my Approach essentially it's the simulated world that's effective for that simulated being right and so for example if you if you depict that simulated being in a world where beings just happen to suffer then you shouldn't feel guilty because you're not causing that that's a bit the idea but as soon as you start talking to the simulated being like and you feed and some data and say look you're actually a simulation here's some photographs of our galaxies and that's what our laws look like and so on then you would at some point overwhelm its Observer state with enough information about our outside world so that whatever we decide to do in our world becomes causally effective for its subjective transitions and then you put should feel guilty at some point yeah so I'm very awake and quick here and it sounds a bit crazy but that's somewhat the predictions that you get right um so probabilistic zombie again to summarize would be a thing where it would have different when you were that thing as compared to how I would see that thing it will this would be different probabilities and that's counter-intuitive because I think that even in in thought experience like you think of like the many worlds interpretation of quantum mechanics or other extreme situations that you think of we typically don't think about it in that along these lines so because that would be a bit strange maybe to finish this one one other thought experiment um that had with George muscle recently so we had a couple of emails exchanged um I I tried to explain the main idea but not sure how well it went um it's a bit the idea again of computer simulation so um in the future you know you run you you buy a lot of computing time and then you simulate on your computer simulated Universe right according to simple laws you have lots of compute so you know you have like Evolution starting and yes and you have a in your simulation and some evolved being comes up and asks about the meaning of it all and what's

going on and you know finds perhaps by experimentation the laws of its simulated world and now you go ahead and say well you intervene in the simulation you change its laws immediately yeah perhaps you know you change something dramatically that's kind of on the sky far away from the simulated being and then at some point that simulated person wakes up and sees in the newspapers that NASA does like astronomers can't believe it suddenly the sky has changed and some regime of the universe what the hell is going on right now that would have a very strange description and approach like mine because in my Approach you can think about the simulated being before reading the newspaper just as long as the simulation is going as usual according to its laws they would say well fair enough you know if you were that simulated being then what you would see is also what we as from from the outside would see right so you and your simulation you will see the simulated sunrise and if you rather simulate like that Observer State that's represented by a simulation you would actually also see the sunrise but at that moment when you change the laws drastically these probabilities would diverge if you were that Observer state that was simulated you would probably see the the old laws just continue right so whereas you looking at your simulation where you intervened you would observe a simulated being with high probability that has actually experienced the change of laws and seen so so one very colorful intuitive way to interpret this would be to say well somehow somehow the the person has leaked out of the simulation that has been replaced by a kind of unlikely change link somehow um no no you have this this other thing that you simulated and you can ask about what would it be like to be that thing right but that would be somewhat a separate question of asking it for this Observer State before the change so you'd have these very strange and very counting to the phenomena of kind of you know persons falling out of it the universes also like which are hard to to visualize I think yeah but it wouldn't happen to you and me because we're not probabilistic zombies relative to each other well yeah but I mean of course you know with all the current um excitement about large language models and giant AI I mean in terms of AI alignment and the ethics of um having large generative models say generative AI in silico uh I think you know I'm just wondering whether some of these arguments are actually practically and morally uh you're in a philosophical sense uh you know um quite prescient you know given current sort of focus on say regulation you know how how expressive how big can these charity AIS come um and to what extent does their training correspond to perceiving receiving observations and trying to make sense of those in a base optimal way using using the principles of universal computation and of course this is all in Silicon usually on a van Newman architecture so I do think that

there is practically possible the possibility of these probabilistic zombies where there's a there's you know perhaps I should rephrase what I have in mind um from from respect from my perspective um as um a modeler and a modeler of modelers you know I.E other people um so you know I can certainly describe my brain or indeed um my family um in terms of uh it looks as if they are um maximizing the evidence for the for the models of their sensed World and then I can try and create a little artifact a little synthetic person and we do this quite a lot in a very very um Elemental way just you know just to simulate toy games and you know economics games and psychophysics experiments with the aim of trying to fiddle with the parameters usually of the prior beliefs of of The Observers uh you know you talk about the laws and the contingencies that are normally written in down in terms of tries in terms of generative models and we build these little synthetic um agents to adjust the prize until their response to an experimental Paradigm matches that of a patient or a subject just to try and get a it's like a digital twin in the computer which you can then now start to ask what would happen if I gave them this synthetic drug or change this connection um or to try and understand the uh the sense making implicit in these so you know there that there is a your very practical um if you like um instance of probabilistic zombies that people are actually using in computational neural Psychiatry for example uh actually at hand as you can see the your um another interesting example of this is the virtual brain project um that's come and deal you spin the auspices of the human brain project and now e-brains um and the whole point of that is is to create digital twins of people with epilepsy so that you can ask well if I did this surgery to relieve this intractable epilepsy what would happen before you actually do it on the actual patient so you know there are people out there building zombies and the reason they're probabilistic zombies perhaps not the virtual brand because that just tries to reproduce neuronal like biophysical Dynamics but but some of the uh agents that we build actually aren't that they actually are with stall in them generative models prior beliefs of A probabilistic Sort that provide the constraints we were talking about before um the other underwrites their constrained maximum entry principle so that we are unlike me trying to describe a member of my family so I can see how my family the member of my family behaves I can look at the Dynamics and I can reverse engineer like kind of model Observer model that that was um if you're like underwriting their Dynamics you know from your point of view that uh you know Universal induction being an expression of some lipunoff function for you it could be sort of you know an algorithmic um so but you would need to I think um at the end of the day write down the model or what you were trying to compress um from you know and from my point

of view your if I was trying to understand a member of my family another another you know real sentient artifact um I I will be reverse engineering their their model their understanding the world here how they were compressing their observations but when I simulate an agent I'm doing I'm doing the other thing I'm doing it in reverse I'm writing down a generative model and then just using the gradient flows and the updates on T to t plus one uh on under the assumption that they are going to um minimize their complexity for example um and this has a name it's called the ideal Bayesian Observer assumption yeah so when you go into biology you always commit to something called the ideal Basin Observer assumption which is just that that your belief updating in your subject simulated or observed um is going to comply with universal computation it's going to be Bayes optimal from the point of view of Bayesian statistician um but of course because you've given it the the generative model which has got nothing to do with the real world you've created a probabilistic zombie and as you say it's probably you know like the model that you've created they are supplying that if you like that objective function of that function to generate a continuous time the Dynamics that you're trying to simulate in a real person in this digital twin for example um because you you're supplying that model it's got nothing to do with what that model actually sees or you know it's not gonna it's got nothing to do with the world out there it's just appropriate for the kinds of inputs and observation supplied that by that particular model um so I think this you know this this notion is it's not just if you like philosophical um a philosophical nicety or a you know an interesting thought experimenter I think it could actually locations you know when you tell your synth your digital twin in fact you're a digital twin uh then they have to do all sorts of belief updating should they want to but you said something quite interesting you said that the digital twin they're my digital twin and the computer they would just carry on observing and and because the way I read it was because they had no way of updating and being able to predict this random fluctuation in the sky and this Bosch part of the sky was bright orange for um they wouldn't be able to see it and you know I'd certainly could go with that you you can't perceive you can't see you can't rep represent you can't compress you can't register anything that's not part of your internal you know repertoire your model of the world so they wouldn't be able to see what was going on even if the person presenting the big orange blob in the sky in in silica or in in you know in the simulation was able to see what the agent was seeing is that what you meant by by you know this dissociation now between the obser The Observer and the Observer of the Observer um I am also wondering a bit so maybe it was a it's actually a bit different so um it's it's nice that you think that there might be actual applications of such

an approach in these scenarios um let me say where I see a potential and then let's see if that's also what you had in mind so I suspect that it there's a it's a bit different though oh I think one question that you may want to ask um at some point is is my language model is my simulated agent you know maybe you simulate a bunch of neurons at some point is somebody at home right so you may want to ask have I created Consciousness that's the way we typically tend to phrase it I think um now we usually think well you know this question is not a question this is essentially a question of the philosophy of mind so you would have to somehow understand what does Consciousness mean make a theory of that and and then try to apply it to that specific structure that's sitting in your computer or so um whereas if something like like my Approach is true then there's a level below and and that's like asking is um if you were that thing would would this be coherent with what the simulation shows right so um and then the answer turns out to be sometimes no which also in particular implies that there's nobody at home in the way that you would intuitively think for Consciousness right even though you didn't say anything about consciousness but the idea is that not being at home here would be so so we tend to say like here's something I can point to something spatially localized perhaps a piece of matter or a particular localized pattern in my and it makes sense to ask what's that thing feeling or what would it be like to be that thing you know um in some sense my Approach denies that it rather says that if you are if you are you yeah whatever that means then you are a pattern right um it will often look as if you were spatially localized and instantiated on some small material part of a larger process like part of a computer simulation or a piece of matter in the world as we think of it but that's not fundamentally the case and since that's not fundamentally the case you can have a Divergence between these two right you will not have a Divergence it will just be the same thing when in in the usual situations like you know in our world we point at stuff that has high complexity or we run a simulation with constant laws point at a persistent pattern there there being that pattern is the same like makes the same probability predictions as we would describe what happens to that pattern but what I described as extremely counter-intuitive you would have a Divergence of these probabilities without being able to point at a mechanistic course of the diversions so if you just you're the simulator you're running your simulation you're changing the laws immediately then in principle there's nothing for you that you can misunderstand yeah you see what the simulation is doing right it's running according to these laws and then the others you see what the simulated being is doing and saying and claiming to see you'll see them watch the newspaper and being surprised suddenly something's going very weird and then it sometimes gets used to

it after a couple of times you can perfectly understand that but when you ask if I were that thing in the simulation shortly before the change happens what what are the chances of what happens to me yeah when these chances are given by the probabilities of the old laws right so um and that's information theoretic reason essentially right it's not um so when I said intuitively this is something like the simulated being falls out of the simulation when I don't have a mechanistic story to tell like something kicked him and he really fell out of it in a mechanistic way you know it was more that the abstract law that predicts what observers see tells you that right and it's just a derived prediction that the first and the third person perspective typically agree so that if I'm the simulator and then I see what my simulator being is experiencing that's what it will actually experience so so the divergence is not mechanistic in the sense it's structural and surprising in the sense very counterintuitive so what you were hinting at I think was was also for me very interesting to learn what you do there but it seemed that you were able to tell a clear story about what happens in that simulation of the like say of your family members of this little agents that have reduced structure that you want to predict what how your family members would respond to treatment you could still say according to the mechanics mechanism of the simulation I can understand this kind of Divergence between the simulated family members and your actual ones whereas I think this would be a different mechanism a different kind of thing this is that just said to answer your question does that make sense nearly yeah it's still uh you know the counter-intuitive and the magic aspect of this is is what's intriguing me here so this notion of being knocked out of the Universe um that I found quite intriguing so are you saying then that the simulated uh Observer from its point of view in the face of some designed or um large fluctuation which was completely unpredictable um so there's no way that that normal um uh sort of Ideal Observer rules apply there's no way that you can um call upon um Universal induction to to probably uh respond to or make sense of this this large fluctuation are you saying then that from the point of view of the Observer um that Observer just disappears gets knocked this uh but what happens to this poor Observer that's seen something that's never seen before in its life and cannot predict and cannot perceive yes so first note the similarity perhaps to the boltzmann brain problem and the way that I tried to ex like come back to the solution is you know proposed here um somehow there you also say well you know um you would expect business on planet as usual to persist regardless of how many copies are out there whereas that will essentially because you are your state and it's the same state here as in the copies so you shouldn't think of us sitting either here or there right you summon all of that at once and now this kind of colorful interpretation also applies to the

simulation yeah you say let's let's interpret this for the moment it's not part of my theory but it's interpret this is the kinds of many worlds picture yeah you can think of all possible worlds one of them is like the world with laws as it evolves like in the beginning of the simulation unchanged forever another world is the simulated world with the change of laws like there are many other possible worlds um where essentially a copy of the very same of server state would appear now the idea is that you are not actually the thing in the simulation but you somehow all of these copies at once it's more like it's just that some of these instantiations have higher statistical weight than others and those that have higher statistical weight are those with a simpler word essentially right so that would be a kind of many worlds picture if you want of why algorithmic complexity or probability would apply here um then it's in this picture you wouldn't really say that you're kicked out of the universe it's just like usually you know you you would think that you're that thing in that regular Universe because that regular Universe has higher weight and you're not actually the other one in the more complex Universe the changing laws now if you're the simulator and you look at the assimilation after changing the laws you will certainly see a simulated being in some Observer state I'm not claiming that this is now a zombie in trauma sense or so no no I mean you could also think of what it would be like to be that thing it's just when you've been that other thing before then that would be a very rare place to end up later right um so it's very weird so maybe you know when you have the simulation and maybe you made friends you think emotionally attached to the simulated being that you had run running around there like and you think like oh I want really that that my friend as well off in the future I don't want to hurt the simulated being and so on and suppose you you want this at the moment before you change the right you want that being to be well off but you also know that the being is very unhappy when it sees a drastic change right maybe that's the personality of the simulated being it wants to be in a predicted to build environment right um but you're now forced to change your laws should you be worried well in some sense no because with with high probability your friend will continue to see regular laws right it just will be elsewhere um you'll then see another being one that has actually seen changed laws and you may make friends with them you may be unhappy that that other thing is unhappy but but you you know um yeah it's it's a bit it's it's even less intuitive than than everettians want to paint the world I think but but it's consistent yeah maybe the many worlds way of thinking framing it I think having helps yeah I think yeah yes I'll have to think about that a bit more I think I'd have to do that with a cigarette and um one practical Point um I have um a lot of friends at the moment in philosophy of Consciousness

um and they are asking um very much like people talk about the neuronal current subconsciousness are there any computational corrects and Consciousness it strikes me on quite a difficult position to to address to address that you know through the lens of a physicist and one one argument that has come to the fore interestingly is can you simulate Consciousness you you almost implies that you know one way of looking at the these uh surprising or counter-intuitive outcomes under the application of this Universal inductive approach um did speak to you know is there anybody home you know in the simulation um one interesting thing that's come out of that is that um it might matter I'm just taking um going back to a theme a conversation we're having before about um the generalized synchronization that ensues when you're in some um open or non-equiv and steady state in exchange with your world as it's going on predictably uh in the right kind of way um it it might matter whether you're doing your simulation on a Von Neumann architecture as opposed to not in memory processing or processing memory uh I think what Jeffrey Hinton refers to as mortal computation so that the actual physics of the Observer State and the Dynamics and the thing that actually has um the complexity um is if you like doing the computation as opposed to simulating the computation um and reading and writing all the laws into memory on a Von Neumann architecture thereby violating if you like or abandoning Landau was bound on the you know on the so she disqualified that for for the viewers and other friends here um that but that one way of looking at that sort of um stepwise change in compression or Universal induction or the movement of a belief structure um with each new observation is you have to raise information to change your mind and in raising information needs a particular amount of thermodynamic work and there's a lower bound on that or an equivalent uh that is known for physics but that only applies when the thing that's changing its mind is the actual thing itself uh I saw this is what I was hoping that Jeffrey intermed by mortal computation that's not what happens in a van in a simulation machine it's not what happens um in uh The Matrix it could be what happens in a brain in a vat if you're talking to a good old-fashioned philosopher but it's not what happens in The Matrix where you've actually got to read and write from memory which takes all as a magnitude more energy just to bring you know to bring stuff out so I was just wondering whether you know it it it's a slightly practical thing but if you um if you now admit to the thermodynamic consequences of your interpretation in a literal sense via things like vigilinski equality and Mandel's principle and you ever really have anybody there even in a simulation if your simulation is not mortal in the sense that it's not using a van human architecture this is a fascinating question I I really like this question because it it kind of opens

up the question like what if you change your model say you make it probabilistic instead of chromograph complexity like which features survive and which change and in particular this question like to the implementation important about how you do the the computation for example I take it I just discussed it with Kevin McQueen about IIT and and what they assume and whatnot like integrated information Theory they we had a long discussion because they seem to insist that the arrangement and the actual physical realization does matter whether you know it can call it think launches or not whereas I thought intuitively I thought that's that's I don't believe that right I think it shouldn't be it should just be about the the pattern itself now two things about that the first thing to say is um the way I formulated it now um the pattern like the way you implement it doesn't matter um well it's a matter of degree I come to that perhaps later but but the second thing to say is that I like the question that you posed because you may ask now if that's my conclusion for this specific model with the bits and so on what if I change it will it then be more dependent on its thermodynamic realization and so on so this is an intriguing question I agree um to say how it's at the moment and the way I formulated it there I would say um in principle every simulation is is just a depiction of an existing structure you're not bringing things into into existence unless you you intervene somehow so let's think about a simulation where I I talk to the simulated being or make some decisions and some interventions and so on and then the complexity of your simulated being is not just the complexity of that pattern or yes it is but when you ask whether however whether you know being that thing is the same as being in our simulation in that world yeah then you have to take the complexity of its description given our words which imply which includes the the location functions or how do I pick out that structure from my world right I have to say where it sits and what kind of computation you have to do to read it out of the computer and if that computation is too complex for example if the simulation is running a very encrypted delocalized way then the readout function might be so much so complex that in total this world with simulation has very low statistical weight for that being right and that it's not actually contained in the simulation so there's some sort of degree um to which like how do how you do your computation you shouldn't make it too contrived and delocalized and so on um yeah that's I think how it how it's at the moment are you happy with that professor friston yes I am yes so we've been talking for a long time now so you'll have to stop us talking about yeah minutes because that is 10 10 old in on the stage it's waiting patiently to ask you both questions so Professor Thurston are you are you happy and satisfied with the questions or do you need more time all of our favorite issues thank you very much let me thank you thank you

for all your time and and it was a lot of fun to talk to you about this so thank you so much it is it is thank you so much so we're going to open up the floor quickly um I have a question but it's not exactly a question I think both of you I have addressed it briefly I'm just having a bone to pick so um recently there's an article about Christoph cough losing a 25 years back on understanding the neurological basis of Consciousness it seems like even with the wine on the line Consciousness remain you know and ethnic wrapped in the brain scan so I just wondered who do you both perceive the instant play of Consciousness and the physical world especially when it comes to the integrated information Theory and Global neural and work placed here with Robinson so Dr Muller could a deep understanding of the physics of the first person perspective have unraveled this mystery at all article that you just mentioned I I was just in Channel we should do all we can to study the neural correlates of Consciousness with all methods of Neuroscience that we have um it's just from the other side from from the philosophy side I see many arguments that there may of like be very hard problems remaining like there's a what trauma is called the pretty hard problem I think that we may solve one day with Neuroscience the heart problem I think we'll need neuroscience and philosophy it will probably need less physics than the physicists would want to believe but that's just my personal guess as like a non-expert thank you Professor first what about your thought about could you send me a link to the paper afterwards and it sounds very interesting so yeah yeah I I want I lost the 25 years to David Chalmer yes this is Christoph is it yeah it is we share the same hero so both Kristoff and myself um have a hero called Sherlock Holmes uh a great Detective from London is a fictional character but that's uh that's one thing we share in common um and a common friendship with Julia turnone who was the architect of IIT integrative information Theory but I wanted to disagree with Marcus I I think we will need physicists to solve this um to be quite honest um and I think it is in the direction of universal computation and these universal laws that basically dissolve any bright lines between information and energy that are going to be at you know the heart of it I also think that um IAT is certainly on the right track but it's it's missing something which Marcus has brought to the table which is the the fact that you know there's a difference between the Observer and The observed and until you think seriously about the relationship and the distinction between the Observer and The observed and the implicit relational aspects and I use a relationship relational as a nod to people like Carla ravelli and relational quantum mechanics um I I don't think there's going to you know I I'm not sure you're going to get a satisfactory answer about Consciousness there has to be something fundamental about the Observer the measurer the

inferior and the and the relationship between that which is being measured and observed and inferred before I think we're going to get any serious uh get seriously close to a philosophical resolution of the hard problem that would be my position and if Marcus wants to push back on on this Pro physicist I still have physics I will not attack the physics professional let's not so we're gonna move on by John have you got you got it you got a question John no I think I'm I'm out of my league here I'm Gonna Leave it to people on the stage okay thank you John I'm going to quickly move on we're going to Adam thank you um I uh I was wondering about the connections uh to the principle of least action um if I understand what you're saying Dr Mueller that uh this principle of universal induction is what's determining is what stands in for you know physical laws and the connection that Dr fishnon you're making to the uh free energy minimization I understand these things in a very rudimentary way but somehow I feel there's a connection to the principle of least action I wonder if you uh might comment on that so so should I start perhaps um because I you your ideas sound somewhat familiar um to something I as a PhD student I wondered about this as well and what like certainly you minimize the action to get the laws Integrations of motion and mechanics um can you understand this as a kind of complexity minimization which is the story you tell from the other direction of thermodynamics and computation and so um in the end my like intermediate like results or conclusion was like perhaps but maybe not that directly as as I thought so so so the way I like to think about the the principle of least action is really in the architecture of the world somewhat of you need a couple of steps to get there and one step below is I think quantum mechanics and and the path integral and when you do an approximation of that which gives you classical physics you end up that you have to minimize the action and once you have something like Quantum there are information theoretic reasons for understanding where the path integral comes from for example and once you're there you can get to the principles of each action so the way to direct the place to look would probably be information theoretic derivations and underpinnings of quantum theory and and the path integral but I think a direct connection with here maybe maybe action is just complexity um among like this and that measure not the other one out of this complexity so I don't think that can be made that easily um yeah for me that there is a direct um a link with the principle of least action I mean you know we often say that the um the free energy principle is just a variational principle of at least action and the action in this instance is just a path integral or a time interval of this bound on marginal likelihood or negative log uh marginal likelihood um so the free energy principle everything that that entails um is a variational principle released action I'm just thinking about

Marcus's um answer so uh I you we we certainly use the pathogable formulation in fact the other most current um and in fact the original formulations the three energy principle um Drew upon the path integral formulation you'd certainly need to use that in contemporary applications of the um of the free energy principle but you don't need the quantum mechanics but you do need the probability distribution so that the way you get the probability densities into play is by associating The Continuous internal bulk or of the or the Observer space states where you are in some continuous Observer phase space with a sufficient statistics or probability distribution so once you do that you're back in a world of um which could which can be cast as Quantum information theory if you if you um and some people have tried to do this um but you don't need to do that you can still work with classical information but now you've basically got back into the world of probably probability densities by associating um the internal states with a problem with parameterizing I represent a probabilistic description of the causes of your OBS of your observations and if you do that then you've got you know you can read the reality principle simply as a as a variational principle of least action I I still submit that you know in the limiting case of um the information encoded by bitstrings um where you you can sort of in the limited case put Delta functions on things and using the laws of large numbers I'm thoroughly hope you can get from that continuous density dynamics that the free energy principle employs to Universal induction um so if if Professor Muller can write the paper showing that they're effective at the same things he can then say and yes this is also a principle of of least action yes thank you a wonderful discussion uh my question is a kind of context based you know if you have an observer in the first person perspective is there something it is contrasted against like an experience of State and they ask that because I think eventually The Observer and the experiencer let's say in quotes are you know information systems they contain some information and like you know what professor frisson said The Observer State face State refers to the least action principle of from coming from free energy so I'm wondering if any one of you both of you or both of you can talk a little bit about this kind of what the Observer is observing with reference to these kind of attributes right some kind of information based and is there a kind of you know can you think of a kind of a database kind of metaphor something relational versus object oriented I just want you to kind of philosophize because I think a lot of people in the audience would love to hear that aspect I want Marcus to go first okay I I'm even less of a philosopher than he is no I'm not sure but the answer will will certainly depend on on the framework and I I feel like I would have to think about it but let me say what I what I understood or what I what intuitively came to my mind

when I heard your question so um the way I think of the Observer state is um just everything that makes up the Observer at a single moment in time which is for you perhaps not the chair over there but it's also not just your experiences it's also memories um including ones that you're not currently thinking about maybe also some aspects that you will never be able to even think about consciously in the future um the point is when the way I formulated my Approach you you can apply it and make predictions and try to understand these Parcels without answering the question what say for me and you this Observer State actually is and how I should interpret it and where in this state sits you know experience versus memory or something like that it's an extremely abstract level it just says somehow basically it says whatever this Observer State means as a bit string um there's no like dictionary that you can open it oh this is not the experience and these zeros and ones are the memory it's more abstract than that the meaning of that status in its relation to other possible bit strengths and the computability structure um it's given to you by information Theory essentially so this is power you can do a lot with it but it's that abstract but it also limits the possibility to answer more concrete questions including your question I don't know the answer to it thank you Marcus Professor Christine yeah um so in contrast um the sort of the classical fep of the person the quantum fep uh formulations uh would actually commit to the notion that the Observer State written as all the inside of something a particle or a person um did um encode beliefs probabilistic beliefs about what caused the thing that's being observed um which means that there is a thermodynamic or a physical instantiation and implementation of the representation and the information as it were about something and the thing it is about is what caused the observations of course you'll never know what caused your observations but you can have beliefs and fantasies about those uh about those and it is um this kind of representation if you like or explanation um to which the complexity minimization applies that you you have the least complex the simplest most parsimonious efficient probabilistic explanation for the causes of your Sensations so I've introduced the word sensation because I mean when we're talking about an observer State one could actually mean the state of the internal um internal aspects or status say of a brain or a thermostat um that is responding to the observations the sensations which would be the you know the Thermo receptor or the uh you know my eyes and my or my sensory epithelia so I think there's a you know there's another delicate set of questions and answers to be had about what what what does an observer State constitute is it the bulk that's on the inside or is it the bolt that's on the inside and the observations that are impressing themselves or represent the way of some putative World talking to the Observer I put it like

that because there's an interesting um talking about sort of uh quantum physics and trying to get us away from philosophy um there's an interesting map between or relationship between um the um the thing that separates my brain from its lived World which are my sensory my sensory organs my sensory epithelia my skin my you know my eyes and the like um uh and the notion of a holographic screen in Quantum information Theory so another way of if you like looking at this sort of um the way that the Observer state is fundamentally separated from what is being observed is to think that um think of it in terms of the separation in my world that would be the classical world that would be a Markov blanket that has these sensory Impressions upon it but in a quantum physics or Quantum information they're not not quantum physics the quantum information theoretic this will be a holographic screen that divides the bulk of one system which could be an observer from another system that is being observed that they can only talk to each other by reading and writing from the screen itself um and on that view um one has to ask is the Observer State the bulk of the Observer or are you allowed to include usually classical information on the holographic screen and if you can or if you can't more interestingly I've of course anybody you know we've spent a lot of time talking about watching my computer simulation and I noticed that you know um Marcus's eyes lit up with all the counter-intuitive exciting implications once we started talking about me looking at another Observer but of course if you've got this holographic principle in play you'll never really know what's going on in the thing you're observing so it's just it will always be as if because there's a you know all you can see of your simulated person or indeed your loved one um is what is projected onto that holographic screen you'll never ever get Beyond it you know you you know the only way you can communicate sort of a classical information at least is is is through that so it wasn't a good answer but you know the experience uh I I'm assuming then is a an attribute of the inside the bulk on the inside the observation is the thing that may live on the Markov blanket or the holographic screen which is your window on the world and possibly that world will contain other observers but that will just be your fantasy yes thank you uh you know for both of you to the wonderful interpretation of that you know I have a follow-up question for for professor fristian because you have in-depth understanding of the psychological the psychiatric side of that same thing so how do you embed the concept of an observer phase State space stability uh is there some you would do you see some kind of link to the psychological aspect of our interpretation of the world and ourselves especially our identity you know I'm just wondering if you can uh you know take us through that for a few minutes right I feel selfish using a few minutes I think we've only got 11 minutes left without

very briefly um that's a great question um and indeed it is why um we committed so much time to investing in in the sort of the physics of sense making and belief updating simply because in Psychiatry and psychology it's all about beliefs and certain psychiatrists when beliefs are slightly aberrant um in relation to uh the thing that's generating the consequences or the things you have to least uh beliefs about and I think you're absolutely right that a a very important aspect of um my compress representation of the um is the representation of me as an actor in that world so you don't have to I mean a thermostat doesn't have to have a sense of self but if I have to engage with other observers it's coming back to this theme which was recurrent in in the conversation with professor you know which was observers of observers if I live in a world that's full of other observers like me um then I'm going to have to have a model of a good regulator like model of a good representation a compressed representation of what it is like to be an observer including myself so I'm going to have self models those models are going to be compressed and what another way of reading the right kind of compression is having the right priors if you remember before the um the statistical view of uh compressibility is putting the prize in the right place so you don't have to move away from them to make to have a good explanation for what's going on including me and how I behave in this situation and how I relate to this other um so people do come along with very very different priors and in a sense you could possibly read all neurology and Psychiatry as people presenting with um buyers that are not quite fit for purpose in their world um and very often they're held with undue Precision or conviction uh which means that they're they're very difficult to shift so you know quite literally they are stuck in a rut but it's a rut of sense making and self-modeling and Universal induction um and then you normally end up asking you know what are the what are the biophysical ways in which we represent that um the Precision of or the commitment to or the confidence in certain representations or beliefs and then there's another story about how that might be um leveraged from the point of view of farmer and uh and Psychotherapy with a cybernetic an AI ecology currently being currently building in the world today how do you anticipate that this will change the subjective observation experience of the individual people used to be able to navigate using Maps but today rely on smartphones with GPS technology and younger people may have lost the skill to navigate using a map how will this technology impact the subject the subjective experience of observing as an amateur and not an expert on AI um it's some aggression that I also find really exciting like how will AI change my everyday life and that of my children in the next five years 10 years 30 years um it's it's very hard to make a concrete prediction like for me especially I'm not an expert but

also when I try to follow the experts who also seem to disagree very much I personally tend to be optimistic so when you talked about navigation so the fact is I was very bad at navigating and finding my way before smartphones now I'm much better even when I turn it off because I feel like when I have it and I make my way for the first time I see the map and I memorize it I get a better understanding and then I can do even better without smartphone so for me it's kind of a win-win thing it's not that I got worse with it I got better with it I like to hope that something similar will happen with other aspects of us and AI that for example we kind of personalized teachers perhaps but tell us in our own best ways of understanding teachers about whatever you want to know and will have to learn um so basically I'm personally optimistic but it's it's not you know I'm not an expert it's really just a personal feeling I'm equally optimistic I I don't really subscribe to these dystopian views um of what might happen if um artificial intelligence or or you know our augmented ability to download certain cognitive capacitors into the environment to use the sort of you know the perspective of Andy Clark I don't really see that you know I have this more utopian view of things um I'll just give you one example I do acknowledge that certain skills will be lost um in children uh basic arithmetic for example but there are other skills that they're gaining you know I'm too old now which is why I asked Marcus to to answer this is as a proper Millennial I imagine uh but I'm too old and I don't have a mobile phone I'm too old to use them because they're too small and I remember being on the tube um about 10 years ago when everybody apart from me was doing sorry the Metro um the underground train um the underground yes the underground uh everybody had their smartphones out apart from me so I could just watch everybody else observe everybody else obvious serving their smartphones now what really struck me was you know children who were able to type using both hands at an incredible speed I mean almost at the speed of an insect that I couldn't physically do so yes they will probably be completely unable to ever repeat their times table but they had such magical flu motive fluency with their thumbs and their ability that was a new skill which you know uh you wouldn't have found it 30 years ago but you certainly won't find in my in my generation so I'm sure it's all going to be okay but things will change in in terms of our competences and the way that we relate to our our world and our e-world so I was just uh I'm just going to go with the uh trained of motion from prakash's uh series of questions and um I was just wondering if um the two of you could touch on um there are a couple of terms that help with this uh observation and it's uh called metacognition or meta awareness so I was wondering if um if one of you or both of you would like to uh to touch upon like metacognition or meta awareness um quick example if I was

reading uh age in a book and say um uh there what happens to some people is they they keep on reading but they lose attention um and they're not processing actually the content that they're reading um consciously um but there are people that uh apparently are aware of when this mind-wandering episode occurs and um the ability uh apparently is men of awareness and this can be cultivated in one's life so I was just wondering if um and it works with memory right um as well and that's why I I'm really happy that you brought up not only just the experience but the memory aspect um so if if one of you would like to touch on uh this will be great and and thank you both very very much I mean I know nothing about this huh perhaps Professor first then I I know lots about this because I've got lots of friends who do metacognition it's a really important and really important thing I have to say that the professor will does know a lot about this because um he was talking about observing The Observer quite a lot here exactly why I thought of it sorry to interrupt but you are absolutely correct and that's exactly what triggered the question within my observation and I think it's very relevant I mean I'm aware that we're out of time now but I think it's an excellent question um because you know not only do we um we we there's a whole science of observing others and observing The Observer and dadic interactions and and theory of mind that speaks to things like autism and the like but of course we are our own uh Observer and we can observe ourselves so as soon as you have that capacity to um decompose in some hierarchical way as it's a hierarchical uh um theory of mind or um high order thought Theory approaches to metacognition um then not only do you have the question of you know what's it like to observe an observer but what happens if that Observer happens to be made I think you get into really deep and interesting questions of a metacognitive sort just to highlight how important that is you know a very sense of self I think um rests upon that medical tip ability you know a thermostat and one could argue possibly a B um and even small pets do not need to know that they are a scented thing in order to operate very very efficiently as uh you know having a basic kind of sentence and an observation um but you know when it comes to very sophisticated creatures like ourselves that have to engage with other things that are equally sophisticated it matters whether you know I can contextualize and infer whether you cause that or whether I cause that and I now need a sense of self in order to have that exchange and to observe and observer that's actually observing me I think at that level you can see how easily things can go wrong uh you know if you don't get these um uh if you don't have the right kind of of self model and again we come back to the potential applications to Psychiatry and you could argue in some instances neurology well so yeah great question uh thank you both of you have given a lot of material

for us to chew on for some time so my question is very simple as free agents The Observers as free agents do they have free will I'd like to know a quick reaction from both of you yes they have free will from my point of view absolutely and in fact from where where Festival has started um you know where am I going to look next I mean you have to select where to look next if you have to do that selection it's only you can do that because you're the only Observer of your Universe then you have to have free will I agree yes you have we will so I'm a compatibilist I like to read Daniel um written one paper about her information Theory can understand your compatibilism perhaps a bit so long story short yes I think we have free will yeah yeah hi thanks this is this is fantastic um uh the thing that piqued my interest was when you mentioned um the emergence of objectivity and I just it was the interesting turn of phrase I thought um and then you went on and you talked about it as like the convergence of probabilities I think um when you're talking about the sunrises and stuff and and sort of like that if an observer calculates the probability and somebody who's observing that Observer calculates the probability and they converged and that is in some sense ability in objectivity um so I guess or something like that first of all did I hear that right and later you went on and said that you don't assume objectivity but you get it as a theorem I think is what you said um so I guess I'm really curious about that does that um is there any dependencies there like does it emerge in time or like in scale like spatial scale or does it does it depend on like the amount of information you know when you get a certain number of bits you you things converge and is that a this convergence is that like a special condition um or is it like a generic thing and then it does are there can do you know are there could there be special opinions were they or it diverges and there's like critical points things go awry and you don't you don't have objectivity or I'm just really curious about that whole thing I'm I'm curious if you can Riff on that a bit yes so that's a great question uh thank you for it so um the the main mathematical background of having such a theorem um has to do with something a bit different but perhaps more well known and that's known as Bayesian coherence so that would describe a situation not in my Approach but just in the usual way that we think the world Works where if two agents that are immersed in say the same environment and that that holds different prior beliefs about the world and now they go along and you know make more and more observations and then under some conditions you can prove that the Prius will essentially converge to the correct distributions if you want or towards each other so the different versions mathematical formulations of that known as Bayesian coherence um also available some versions are in quantum theory right um the interpretation of this would really be just epistemic about

knowledge I say here's a guy with beliefs another person with beliefs the more they learn the more they agree and you cannot agree to disagree ultimately so that's what what will happen and what I'm doing um you don't have that available so you don't have a joint environment into which you would be fundamentally actually immersed still like the the mathematical reason why I get such a theorem is essentially the same and then from the usual formulation you can kind of learn what what is important there it's not so much the scale like how large The Observer is of the matter or the time it's more the amount of information that they hold um the idea is that you as an observer um observe this emerging world and that's something that that's predicted by this approach with certain probabilities so suppose that's the case then you point at some thing some structure in your world yeah um and that evolves in time as well and that's what allows you to compare probabilities your probability of how that thing evolves versus the probability if you were that thing right and the probabilities are close if the information content of the thing that you're pointed to is large right so that's essentially again the Criterion of that not being a probabilistic zombie so that's what I called earlier so essentially the information content of what you're pointing to of the other Observer must be larger than the complexity of the probabilistic loss that governs its uh it's the relevant features of the world so the more so it's like asymptotically it's in the limit the theorem tells you in the limit of learning infinitely much information um your and the first your third person probabilities and that things first person probabilities will converge towards each other just that sounds very much like a sort metacognitive or perhaps uh Bayesian coherence version of the law of requisite variety uh where the variety is just the degrees of freedom available to hold that information uh in uh I have to look that up I don't even know what that is I make lots of notes here and stuff to read up for me it's it's from the same stable of cyberneticists that did the good regulator theorem but you know the subsequently developed into the lower lower of requisite variety which is just having enough degrees of freedom in the thing you're pointing at is commensurate with the the informational complexity or you know of the thing that's doing the pointing I imagine anyway that's that's your job I don't want to spend uh West anymore anybody's time uh um because I you know that was a great answer it's uh hits you in the face when you uh certainly uh model communication so when the thing you're observing is actually being generated by another Observer by an IQ uh them very very quickly this sort of um requirement that neither of you can be probabilistic zombies and you actually have to have to convince that generalized synchronization uh see I am the cause of your observations and your observations that cause them uh of the cause of my observation are you are the cause of my

observations you get you get some you know really nice mathematical image of the ensuing Bayesian coherence and the convergence on it um on a um a synchronization manifold uh to use the jargon of generalized synchrony uh that it is just that I just asked professional would this be um akin to a principle of unitarity and entangling in quantum physics everything basically ultimately coheres is that the idea um maybe it's it's the opposite in a way we have Howard Wise Men here visit and talk about weakness friend in fact where the whole puzzle comes from unitarity you have the friend in the box and Wigner on the outside Wigner describes the box and the friend unitarily the friend describes it as a measurement so so in a way there you would have a situation where you specifically don't have the communication and then you don't know what to do then you wouldn't have because since the consistency of viewpoints and you don't know what to say so so that goes a bit back to your earlier comment which I wholeheartedly agree so as soon as you have communication between two agents then this kind of um gets rid of zombie status right and um you have to have synchronization somehow of your probabilities either in an epistemic way whether you would usually interpret it or in my Approach really as chances of what happens to you see this whole concept of Observer is very interesting so my own Insight is that the Observer cannot be modeled whereas Observer state is I think what we are talking about but the Observer itself can not be modeled because of the very nature of Observer just as for example The Taste and other things are a model of reality but you know of course scientists who may say that it is like you know the the content of the chemical action in the brain or something like that but that is just equivalence you know as I experience it exists independently in that sense like The Observer exists independently in reality it seems like I just wanted to get your views but if you do consider that observers cannot be modeled and it's not a thing because if it has a thing given that would be observed basically that's the point and uh and it always you know the moment you try to observe it then the Observer has gone back you know meaning like you try to model something and it has gone prior to that that state basically so in that sense it forever remains independent of whatever you try to do uh so these are some of my insights and of course we have a totally different philosophy based on it called advaita and other things just for information so you are very much uh you know actually modeling assuming this type of reality it seems like to me and I'm quite fascinated with it and they're going to study more in more detail okay thank you I should quickly answer to that I'm probably running out of time but just briefly so thanks for the question indeed you're pointing perhaps a two different things like two aspects of it one is like difference between our server States and observer um this is something that's common not

only to my Approach but also to other philosophical questions about personal identity you know in splitting scenarios in aberrating many worlds and so on you know the tele Transportation Paradox where one person becomes two or so turns out that in these situations you cannot meaningfully talk about a person and identity of persons in a way that we used to in language right um so that's that's one observation the other question that you're touching at is just the distinction between the way we model things mathematically and what we think we really are right so my claim here is in some sense we we're more like a pattern rather than a piece of matter but but even when we talk about these Observer States as some pattern you can ask am I the pattern or is the pattern describing what I am or how should I think of this relation and this isomorphism and fortunately I'm ultimately a physicist's don't have to answer because I cannot I think this is a deep philosophical Channel or question I would guess thank you both of you for being here today and it has been a wonderful wonderful discussion and it's a lot to unpack today Professor Christine I will send you the link of uh Kristoff the 25 years the article will send you after the the podcast so this is it gentlemen thank you so much and it's brilliant to have both of you on I would like to extend the time can everybody unmute your mind and wish things before both gentlemen leaving the stage can you all unmute and say thank you please thank yes thanks a couple of Sherlocks up there thank you so much remember to see you all together thanks for the conversation a lot because that was wonderful bye-bye have a good night